

SHOT3R: STREAMING MULTI-PERSON 4D RECONSTRUCTION FROM MULTI-SHOT VIDEOS BY DECOUPLING ALIGNMENT AND RECURRENT STATE PROPAGATION

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Figure 1: Shot3R streams successive monocular shots into a common world frame while preserving camera, motion, and identity continuity. Angles denote source-camera rotations from the first shot.

ABSTRACT

Reconstructing human motion in a common world coordinate system from a monocular video composed of multiple shots is challenging, as abrupt shot transitions can cause drastic changes in viewpoint, visible scene content, and human appearance. Existing online methods generally assume temporally continuous input, while prior multi-shot approaches jointly process shots or refine the observed sequence rather than producing recurrent estimates as frames arrive. At a shot transition, recurrent history provides a reference for recovering the inter-shot world relation, yet propagating the same state through the new shot can induce time-varying camera drift, while reinitialization alone loses that reference. To address this conflict, we present SHOT3R, an online framework for streaming multi-person 4D reconstruction from multi-shot videos that decouples inter-shot alignment from recurrent state propagation. At each boundary, a history-conditioned path estimates the inter-shot transform, while an independently reinitialized path supplies the state propagated through the new shot. Geometry-based association establishes cross-shot identities, and a shared transform registers cameras and humans to the existing world frame. Across three multi-person benchmarks, SHOT3R reduces the equally weighted mean W-MPJPE by 9.5% and improves mean IDF1 by 0.099 in absolute terms over direct recurrent-state carry-over. On the predefined extreme-viewpoint subset of EgoBody, SHOT3R reduces W-MPJPE by 29.9% while retaining frame-wise streaming inference without future frames or per-video optimization.

1 INTRODUCTION

Reconstructing globally consistent multi-person motion from a monocular video composed of multiple shots requires connecting human trajectories across abrupt camera changes (Pavlakos et al., 2022b; Zhang et al., 2025b; On et al., 2026). A shot transition breaks camera-motion continuity and can substantially alter human appearance, leaving individual-shot reconstructions in incompatible world frames and fragmenting person identities.

Here, *multi-shot* denotes temporally successive segments of a monocular video. SHOT3R receives one frame at a time and never jointly observes synchronized views of the same scene state.

Prior multi-shot methods typically process adjacent shots jointly or refine an observed sequence (Zhang et al., 2025b; On et al., 2026). They therefore do not recurrently produce estimates as frames arrive. Streaming reconstruction instead processes inputs in order, maintains history in a fixed-size state, and avoids per-video global optimization, but existing formulations generally assume continuous observations (Wang et al., 2025b; Chen et al., 2026b; Zhao et al., 2026).

Shot transitions thus expose conflicting effects of recurrent history. History provides a reference for inter-shot alignment, but propagating the same state can induce within-shot camera drift; reinitialization avoids this carry-over but loses the cross-shot reference. Our key observation is that alignment and subsequent recurrence should therefore be treated as separate decisions: the preceding state may be read temporarily to estimate the inter-shot relation, while an independently reinitialized state drives reconstruction in the new shot. A same-checkpoint control confirms that state carry-over induces time-varying camera drift within the new shot, beyond a fixed coordinate offset (Section 4.3).

We introduce SHOT3R to realize this separation. At a shot boundary, a history-conditioned alignment path uses the preceding state and current image to estimate the inter-shot transform, then discards its temporary state. An independently reinitialized path supplies the only state propagated forward. Geometry-based person association links identities, and one shared transform maps cameras and humans into the existing world frame. The procedure accesses no future frame, revises no emitted result, and performs no per-video optimization.

Experiments across multiple multi-person benchmarks show that SHOT3R improves world-frame human reconstruction, camera consistency, and identity continuity while retaining streaming inference, with larger gains under substantial viewpoint changes.

Our contributions are:

- We identify and empirically characterize the conflicting roles of recurrent history at shot transitions: pre-transition history provides an alignment reference, propagating the same state can induce within-shot camera drift, and reinitialization alone loses the cross-shot reference.
- We realize this separation with a temporary history-conditioned alignment path and an independently reinitialized recurrent path, supported by geometry-based association and a shared camera-human transform.
- We improve world-frame human reconstruction, camera consistency, and identity continuity across multi-person benchmarks, with larger gains under substantial viewpoint changes and no future frames or per-video optimization.

2 RELATED WORK

World-coordinate human reconstruction. Monocular human mesh recovery spans image-based regression (Kanazawa et al., 2018), temporal modeling (Kocabas et al., 2020; Goel et al., 2023), and full-frame multi-person estimation (Sun et al., 2021b; 2022; 2023; Baradel et al., 2024; Wang et al., 2025c; Sun et al., 2024). Building on these advances, world-coordinate methods recover global trajectories through motion priors and sequence optimization (Yuan et al., 2022; Ye et al., 2023; Kocabas et al., 2024; Li et al., 2024), joint human-camera modeling (Wang et al., 2024b; Shin et al., 2024; Shen et al., 2024; Yin et al., 2024), or perspective and scale calibration (Patel & Black, 2025; Yang et al., 2025). More recent systems jointly reconstruct human and scene geometry (Müller et al., 2025; Liu et al., 2026b; Zhang et al., 2025c; Li et al., 2026a), while OnlineHMR studies

online world-coordinate recovery (Zhao et al., 2026). These methods primarily assume temporally continuous video; we study spatial and identity consistency when a shot transition invalidates that continuity.

Stateful online reconstruction. Online geometry spans SLAM, learned dense mapping, and feed-forward point-map models (Teed & Deng, 2021; Sun et al., 2021a; Wang et al., 2024a; Leroy et al., 2024; Zhang et al., 2025a; Wang et al., 2025a). Streaming methods retain history through spatial memory, progressive registration, or a persistent recurrent state (Wang & Agapito, 2025; Liu et al., 2025; Wang et al., 2025b). HUMAN3R unifies cameras, scenes, and multiple people in the stateful formulation; later work extends contact, representation, memory, dynamic geometry, and state adaptation (Chen et al., 2026b; Sur et al., 2026; Abe et al., 2026; Chen et al., 2025; Wu et al., 2025; Zhuo et al., 2025; Chen et al., 2026a; Zheng et al., 2026; Li et al., 2026b). These methods retain history under continuous observations. SHOT3R instead uses incompatible pre-transition history for alignment without propagating it through the new shot.

Multi-shot human reconstruction. Multi-shot reconstruction seeks consistency across sequential monocular segments, unlike synchronized multi-view reconstruction. Directly related work studies TV-series tracking and re-identification, cross-shot human recovery, actor–environment reconstruction, global motion, and multi-person mesh tracking (Pavlakos et al., 2022a;b;c; Zhang et al., 2025b; On et al., 2026; Kim et al., 2025). Identity association also relates to 3D-aware tracking and appearance- or keypoint-based re-identification (Rajasegaran et al., 2022; He et al., 2021; Yuan et al., 2025; Somers et al., 2024); ours uses predicted 3D geometry at the first post-transition frame. Synchronized multi-view methods instead fuse simultaneously available views (Rojas et al., 2025; Yang et al., 2026; Liu et al., 2026a). Existing multi-shot systems operate over shot pairs or observed sequences; SHOT3R processes one frame at a time and separates inter-shot alignment from recurrent state propagation.

3 METHOD

3.1 PROBLEM FORMULATION AND OVERVIEW

Given a monocular stream $\{I_t\}_{t=1}^T$ with shot transitions, SHOT3R predicts camera pose C_t and multiple humans $\mathcal{H}_t$ in a common world frame while maintaining a recurrent state S_t . Within a shot, the state supplies a stable spatiotemporal reference. At a transition, the preceding state can provide a reference for relating the two shots. Propagating it further, however, may introduce camera drift within the new shot, whereas reinitialization alone discards the cross-shot reference.

An image encoder maps I_t to tokens F_t , and detected people are represented by prompts H_t . Together with an initial camera token p_0 , they interact with the previous state through decoder D_θ :

$$[F'_t, p_t, H'_t], S_t = D_\theta([F_t, p_0, H_t], S_{t-1}). \quad (1)$$

The updated tokens predict a camera-local scene point map P_t^c , a camera-to-world pose, and the SMPL-X parameters of every visible person; its world-space scene map is $P_t^w = C_t P_t^c$ in homogeneous coordinates.

We abbreviate this online mapping as

$$(y_t, S_t) = f_\theta(I_t, S_{t-1}), \quad y_t = (C_t, P_t^w, \{J_t^i, V_t^i, q_t^i\}_{i=1}^{N_t}), \quad (2)$$

where J_t^i , V_t^i , and q_t^i are the SMPL-X (Pavlakos et al., 2019) joints, mesh vertices, and anonymous slot of person i . We instantiate f_θ with pretrained HUMAN3R, whose state jointly represents cameras, scene geometry, and people (Chen et al., 2026b). Our concern is the transition required when I_t is no longer a continuous observation of the stream encoded by S_{t-1} .

SHOT3R separates inter-shot alignment from recurrent state propagation. At a detected boundary b , the same frame is evaluated with the preceding state to recover the inter-shot relation and from an empty state to initialize subsequent recurrence:

$$(\tilde{y}_b, \tilde{S}_b) = f_{\theta, \phi}(I_b, S_{b-1}), \quad (y_b^r, S_b^r) = f_\theta(I_b, S_\emptyset). \quad (3)$$

Only S_b^r is propagated beyond the boundary. Person association and a shared transform then preserve world-frame and identity consistency (Figure 2).

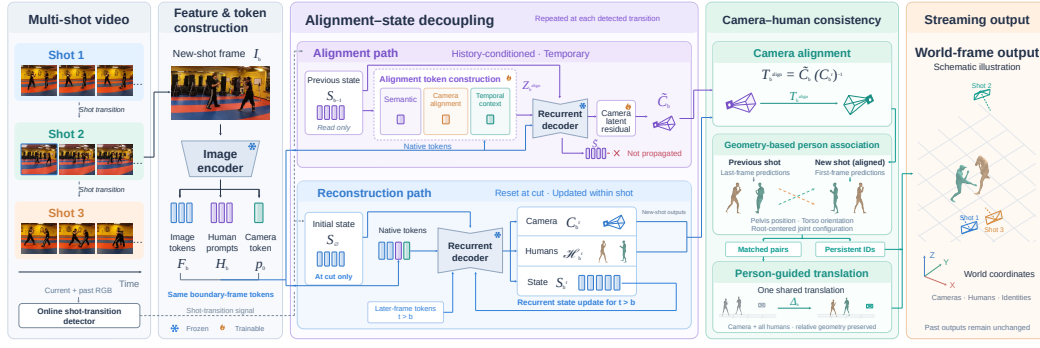


Figure 2: Overview of SHOT3R. The history-conditioned path estimates inter-shot alignment, while only the reinitialized path propagates recurrent state. Geometry-based cross-shot person association links identities, and a shared transform maps camera, human, and world-space scene outputs into the existing world frame. The rightmost panel is schematic.

3.2 HISTORY-CONDITIONED SHOT ALIGNMENT

An online transition detector identifies the first frame b of a new shot from current and past RGB only. It determines when alignment is invoked but does not estimate the spatial transform or person correspondence. No future frame, camera calibration, or body annotation is available to the detector; architecture and trigger details are in Appendix B.

At the detected boundary, the shot-alignment module Φ_ϕ relates the current observation to the preceding state. It keeps three sources of evidence distinct through semantic, camera-alignment, and temporal-context tokens. They encode current-to-history visual evidence, latent camera change, and preceding correction context, respectively, and are decoded jointly with the frozen backbone tokens. Attention pooling first gives current evidence c_b and preceding memory m_{b-1} , from which

$$\begin{aligned} z_b^{\text{sem}} &= \alpha_b c_b + (1 - \alpha_b) m_{b-1}, & z_b^{\text{cam}} &= \text{MLP}_{\text{cam}}[p_b, p_{b-1}, \Delta p_b, m_{b-1}], \\ z_b^{\text{temp}} &= \text{MLP}_{\text{temp}}[\bar{z}_{b-1}, \delta_{b-1}, g_{b-1}], & Z_b^{\text{align}} &= [z_b^{\text{sem}}, z_b^{\text{cam}}, z_b^{\text{temp}}]. \end{aligned} \quad (4)$$

Here α_b mixes current and remembered semantic evidence; p , $\bar{z}$, δ , and g denote camera tokens, the preceding alignment summary, camera latent residual, and its gate. Together with the human latent-residual branch, these representations adapt the boundary prediction to the existing world reference and yield a temporary camera estimate $\tilde{C}_b$. Appendix C details the pooling and residual branches. We learn Φ_ϕ from four-frame dynamic-human sequences in a shared world frame. Transition clips follow $A(t), A(t+1), B(t+2), B(t+3)$; continuous clips contain four frames from one camera. The former supervise inter-shot changes, and the latter discourage corrections under continuous input.

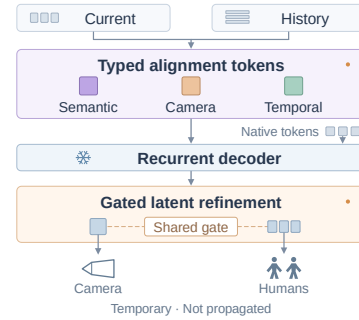


Figure 3: **Shot-alignment module.** Typed tokens relate current and historical evidence; gated refinements are not propagated.

The alignment objective is

$$\mathcal{L}_{\text{align}} = \mathcal{L}_t + 5\mathcal{L}_R + 10\mathcal{L}_h + 0.05(\mathcal{L}_g + \mathcal{L}_{\text{imp}}) + 10^{-5}(\mathcal{R}_{\text{cam}} + \mathcal{R}_{\text{human}}), \quad (5)$$

where $\mathcal{L}_t$ and $\mathcal{L}_h$ are Smooth-L1 camera and human translation losses, and $\mathcal{L}_R$ is quaternion geodesic distance. Let e and e^{raw} be normalized corrected and unmodified camera errors. The gate and improvement terms are $\mathcal{L}_g = \text{BCE}(g, \text{clip}(e^{\text{raw}}, 0, 1))$ and $\mathcal{L}_{\text{imp}} = [e - e^{\text{raw}}]_+$.

Only the typed tokens, camera and human latent-residual branches, and low-rank head adapters (Hu et al., 2022) are trained: 19.62M parameters, or 1.63% of the model. The image encoder, recurrent

backbone, base prediction heads, and Multi-HMR component (Baradel et al., 2024) remain frozen. These parameters sit at the interface between history and the current observation, learning how to relate shots without relearning within-shot reconstruction.

3.3 SEPARATING ALIGNMENT FROM STATE PROPAGATION

The two boundary evaluations in Eq. 3 have different information lifetimes. The history-conditioned state $\tilde{S}_b$ is temporary, whereas subsequent new-shot frames use only the reinitialized path:

$$(y_t^r, S_t^r) = f_\theta(I_t, S_{t-1}^r), \quad t > b. \quad (6)$$

Thus history can establish an explicit spatial relation at the boundary but has no recurrent path into later new-shot states.

The reinitialized path predicts C_b^r from the same image. Since the paths share the observation but differ in historical conditioning, their relative camera transform aligns the new shot to the existing world frame:

$$T_b^{\text{align}} = \tilde{C}_b(C_b^r)^{-1}. \quad (7)$$

After forming this transform, all temporary alignment-path predictions and $\tilde{S}_b$ are discarded. The operation repeats independently at later transitions and never revises previous outputs.

3.4 CAMERA AND HUMAN CONSISTENCY ACROSS SHOTS

T_b^{align} first maps reinitialized camera and human predictions into the preceding world frame. Because camera alignment alone does not identify people, SHOT3R matches them one-to-one using predicted pelvis position, torso orientation, and root-centered joint configuration:

$$d_{ij} = \hat{d}_{\text{root}}(i, j) + \hat{d}_{\text{torso}}(i, j) + \hat{d}_{\text{joint}}(i, j). \quad (8)$$

The three costs are median-normalized before Hungarian assignment (Munkres, 1957). Matching uses no identity labels, camera calibration, or future frames; unmatched detections receive new anonymous identities.

For matched pelvis locations r_{b-1}^i and aligned new-shot locations $\bar{r}_b^j$, a robust shared translation is

$$\Delta_b = \lambda \text{median}_{(i,j) \in \mathcal{M}_b} (r_{b-1}^i - \bar{r}_b^j), \quad \lambda = 0.5. \quad (9)$$

The fixed λ interpolates between camera-only and full person-derived translation and is unchanged across datasets. The alignment-path camera fixes orientation and the initial world relation, while matched-person displacement adjusts their shared translation without changing camera-human relative geometry.

With $T_b^\Delta = \begin{bmatrix} I_3 & \Delta_b \\ \mathbf{0}^\top & 1 \end{bmatrix}$ and $T_b = T_b^\Delta T_b^{\text{align}}$, the outputs are

$$C_t' = T_b C_t^r, \quad P_t'^w = T_b P_t^{r,w}, \quad J_t'^i = T_b J_t^{r,i}, \quad V_t'^i = T_b V_t^{r,i}, \quad q_t'^i = \pi_b(q_t^{r,i}), \quad S_t' = S_t^r. \quad (10)$$

Hence camera, world-space scene points, and humans share one world transform, whereas the camera-local point map is not deformed and the recurrent state is neither transformed nor copied from the previous shot. Empty-match handling and coordinate conventions are given in Appendix C. At later boundaries, the same operation estimates a new transform relative to the accumulated world frame. Transform composition affects only the current and future outputs; already emitted frames remain unchanged.

4 EXPERIMENTS

We evaluate multi-shot multi-person reconstruction in a common world frame, with emphasis on large viewpoint changes, alignment and state propagation, person association, transition detection, and efficiency.

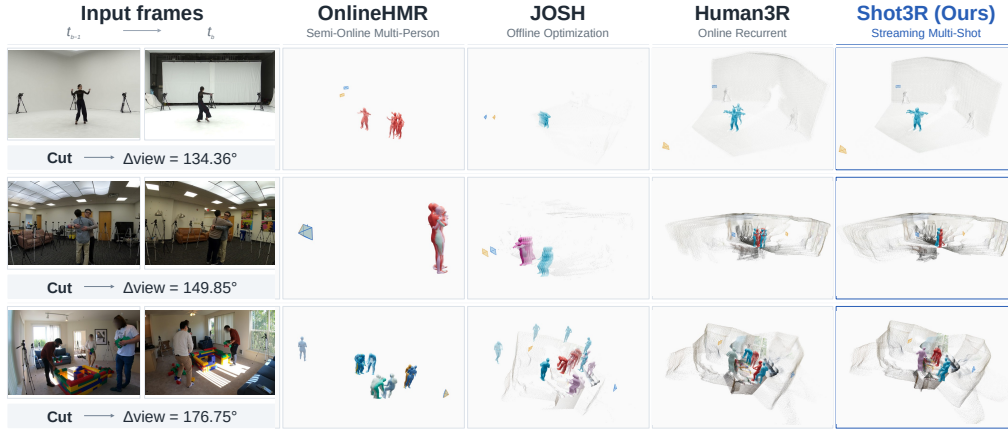


Figure 4: **Key qualitative comparisons across shot transitions.** Inputs before and after each transition are followed by OnlineHMR, JOSH, HUMAN3R, and SHOT3R. Viewpoint changes are 134.36°, 149.85°, and 176.75° from top to bottom. SHOT3R maintains a more coherent camera-human layout; scene geometry is shown where available. The complete seven-method view is in Appendix D.6.

4.1 EXPERIMENTAL SETUP

Baselines. HUMAN3R is the primary controlled baseline: both methods receive the same ordered RGB frames and evaluation, but HUMAN3R propagates state uninterruptedly whereas SHOT3R aligns the shots and propagates a reinitialized state through the new shot. We also run the official implementations of OnlineHMR, TRACE, PromptHMR, JOSH, and TRAM (Zhao et al., 2026; Sun et al., 2023; Wang et al., 2025c; Liu et al., 2026b; Wang et al., 2024b). Table 1 groups them by temporal access and marks explicit transition handling; preprocessing and output scope are reported in Appendix A. OnlineHMR combines semi-online person tracking with a separate global-camera backend, PromptHMR and JOSH process the complete video, and TRAM is included in the standardized full-system timing study. HumanMM and Multi-THuMBS lack the complete public code, weights, and configurations needed for our fixed inputs; their task scope is summarized in Table 15 rather than mixing results from different protocols.

Datasets and metrics. We convert synchronized EgoBody, EgoHumans, and Harmony4D captures (Zhang et al., 2022; Khirondkar et al., 2023; 2024) into two sequential monocular shots, yielding 129, 90, and 88 fixed cases, respectively. The model receives one temporally ordered frame at a time. The datasets provide complementary evidence: EgoBody supports controlled mechanism ablations, EgoHumans emphasizes large viewpoint changes and recurrent-state carry-over, and Harmony4D evaluates person association and repeated transitions.

W-MPJPE (W) fixes a Sim(3) alignment from the earliest valid observations and is sensitive to cross-shot displacement; WA-MPJPE (WA) aligns all valid observations. Camera error is ATE-Sim3 on EgoBody/Harmony4D and ATE-SE3 on EgoHumans. IDF1 measures identity continuity (Ristani et al., 2016), and Coverage exposes missing predictions omitted by conditional human errors. Absolute errors are compared only within a dataset; full definitions and aggregation are in Appendix A.

Implementation details. We train the 19.62M alignment parameters on AvatarReX and THuman while freezing the remaining model. Each source contributes 8,000 transition and 2,000 continuous four-frame clips, sampled with probabilities 0.6/0.4. AdamW (Loshchilov & Hutter, 2019) runs for 7,200 updates (batch size 10; resolution 512). Every evaluation uses the epoch-72 checkpoint, $\lambda = 0.5$, and one inference configuration, without dataset- or video-specific optimization. Appendix B gives the data composition, optimization schedule, and training curves.

Table 1: Comparison on the EgoBody and EgoHumans multi-shot multi-person reconstruction protocols.

Category	Method	Transition	EgoBody ($N = 129$)					EgoHumans ($N = 90$)				
			W ↓	WA ↓	ATE ↓	IDF1 ↑	Cov. ↑	W ↓	WA ↓	ATE ↓	IDF1 ↑	Cov. ↑
Offline	PromptHMR (SPEC)	×	741.4 ⁶⁴	336.1 ⁸⁷	—	0.195	0.202	1193.3 ¹⁷	487.7 ²⁵	—	0.061	0.062
	JOSH	×	399.6 ¹²⁷	322.9 ¹²⁷	0.087 ¹²⁷	0.973	0.980	1410.7 ⁷⁷	481.3 ⁷⁷	1.755 ⁸⁷	0.616	0.686
Semi-online	OnlineHMR	×	961.2 ⁹⁷	418.1 ⁹⁷	0.259 ¹²⁹	0.375	0.424	1311.7 ⁵⁰	405.5 ⁵⁰	3.869 ⁸⁷	0.142	0.211
Online	TRACE	×	576.4 ²²	276.4 ³⁵	—	0.036	0.035	2464.6 ⁴³	749.5 ⁵⁹	—	0.070	0.072
	HUMAN3R	×	321.1	233.4	0.124	0.870	0.981	1005.1	365.7	1.673	0.471	<u>0.639</u>
	SHOT3R	✓	258.8	201.7	0.016	0.985	0.981	866.8	336.9	1.423	<u>0.571</u>	0.616

Transition denotes explicit shot-transition handling. W/WA are in mm and ATE in m (Sim(3) on EgoBody; SE(3) on EgoHumans). Superscripts give the number of evaluable streams for conditional geometry metrics; IDF1 and Coverage use the complete predefined protocol. Geometry means computed on different supports do not constitute a common ranking.

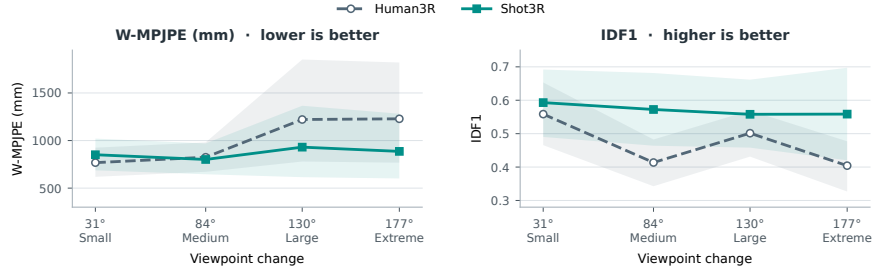


Figure 5: **Viewpoint response on EgoHumans.** Curves report absolute W-MPJPE and IDF1 in four fixed angle strata; bands are capture-bootstrap 95% confidence intervals. SHOT3R improves IDF1 throughout, with the clearest W-MPJPE reductions at 130° and 177°.

4.2 MULTI-SHOT MULTI-PERSON RECONSTRUCTION

Overall results. Table 1 reports EgoBody and EgoHumans; the complete Harmony4D comparison is in Appendix D. SHOT3R improves WA, camera error, and IDF1 over HUMAN3R on all three datasets. On EgoBody, W/WA falls from 321.1/233.4 to 258.8/201.7 mm, ATE from 0.124 to 0.016 m, and IDF1 rises from 0.870 to 0.985 at unchanged 0.981 Coverage. Local pose and mesh accuracy remain essentially unchanged, indicating that the gains arise primarily from world-frame registration, camera estimation, and identity continuity rather than from relearning within-shot body reconstruction.

Figure 4 shows transitions of 134.36°, 149.85°, and 176.75°. Despite pronounced changes in viewpoint and appearance, SHOT3R retains a more coherent camera–human layout; scene geometry is visualized where the corresponding method produces it.

Across all three datasets, SHOT3R improves WA, camera trajectory accuracy, and IDF1 over HUMAN3R. W also improves on EgoBody and EgoHumans, while Harmony4D exhibits a modest increase; complete per-dataset results and boundary diagnostics are provided in the supplement.

Large viewpoint changes. With equal weight per dataset, SHOT3R reduces W/WA by 9.5%/8.8% over all views and improves IDF1 by 0.099. The improvements are more pronounced on the predefined large-viewpoint subsets, reaching 19.1%/18.7% and 0.195. On extreme-view EgoBody, W falls from 505.1 to 354.3 mm (29.9%) and IDF1 rises from 0.758 to 0.983. The supplement reports the complete per-dataset strata and continuous-angle analysis; these results do not establish monotonic improvement for every metric.

Comparison with public methods. Against OnlineHMR on shared valid inputs, SHOT3R lowers human and camera errors and, over all streams, raises IDF1/Coverage; paired intervals exclude zero except for EgoHumans WA. On streams evaluable by both methods, it also lowers W/WA/ATE versus the offline JOSH pipeline across all three datasets, without future-frame access. TRACE and

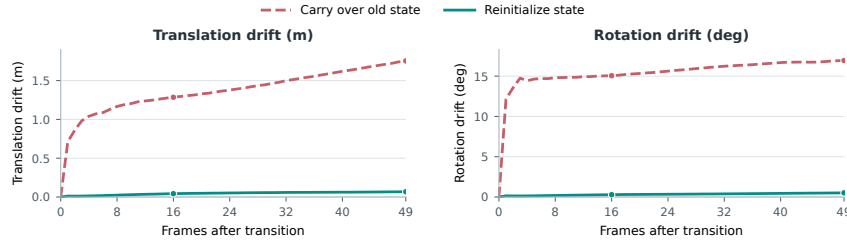


Figure 6: **Old-state carry-over causes within-shot camera drift.** On static-camera captures, the same unmodified HUMAN3R checkpoint processes identical new-shot RGB with no SHOT3R module. First-prediction anchoring removes any fixed transform; carry-over then drifts while reinitialization remains stable.

Table 2: Alignment–state-propagation 2×2 control and the complete SHOT3R on EgoBody.

Configuration	Boundary cue	Shot-transition operation			EgoBody metrics			
		Alignment	Propagated state	Association	W (mm) ↓	WA (mm) ↓	ATE (m) ↓	IDF1 ↑
HUMAN3R	—	×	Continued	×	321.1	233.4	0.124	0.870
State reinitialization	Annotated	×	Reinitialized	×	752.6	589.4	1.141	0.834
Alignment + continued state	Annotated	✓	Continued	×	293.3	227.3	0.082	0.868
Alignment + reinitialized state	Annotated	✓	Reinitialized	×	352.8	267.7	0.017	0.834
SHOT3R	Detected	✓	Reinitialized	✓	258.8	201.7	0.016	0.985

Annotated boundaries are used only by the controlled configurations; HUMAN3R receives no boundary cue. The two aligned controls disable association and person-guided translation, so only the state propagated after alignment changes.

PromptHMR return EgoBody W/WA for only 22/35 and 64/87 streams; Appendix D reports common support, paired intervals, failures, and complete metrics.

4.3 ANALYSIS AND ABLATION STUDIES

Alignment and state propagation. Reinitialization alone loses the inter-shot reference, increasing EgoBody W/WA from 321.1/233.4 to 752.6/589.4 mm. To isolate state interference, we use the same original HUMAN3R checkpoint on identical post-transition frames in 90 EgoHumans cases with a static physical camera and no SHOT3R module. Both trajectories are anchored to their first new-shot prediction, removing any fixed coordinate transform. Over the next 16 frames, continuation produces 1.124 m/14.58° camera drift, versus 0.026 m/0.20° after reinitialization, with the latter lower on all 27 captures. Paired mean differences are 1.098 m [0.832, 1.372] and 14.38° [8.92°, 20.49°], with $p = 7.45 \times 10^{-9}$ for both one-sided paired tests. The preceding state therefore induces time-varying within-shot error, not merely a fixed coordinate offset; full statistics are in Appendix D.4.1.

Under the same annotated boundary and learned alignment, propagating the history-conditioned versus reinitialized state gives identical first-frame alignment error. With association and person-guided translation disabled, continued state benefits world-frame human estimates (293.3/227.3 versus 352.8/267.7 mm W/WA), whereas reinitialization improves sequence ATE (0.017 versus 0.082 m); the 43-recording paired ATE difference is 0.065 m [0.049, 0.085]. Thus neither state choice is uniformly better. The complete system combines reinitialized recurrence, association, and the shared transform, reaching 258.8/201.7 mm W/WA, 0.016 m ATE, and 0.985 IDF1—the best joint outcome in Table 2. Because this row evaluates the combined design, its gains are not attributed to state choice alone. Together, the controls support treating alignment and recurrent-state propagation as separate decisions.

Person association. On Harmony4D, pelvis position already provides effective association. Augmenting it with torso orientation and root-centered joints raises identity continuation from 78.41% to 80.68% and IDF1 from 0.620 to 0.635, supporting the complementarity of the predicted geometric cues. Complete denominators and results are reported in Appendix D.5.2; association remains a lightweight supporting mechanism after inter-shot alignment.

Repeated shot transitions. A small three-shot control in the supplement verifies that the same transition operation can be reapplied, while retaining the associated metric trade-offs (Appendix D.5.3).

Transition detection and efficiency. The detector’s first trigger matches the annotated transition in every primary protocol stream. Under the standardized reconstruction protocol, timing includes model inference, output decoding, person association, and world-coordinate registration, but excludes model loading, RGB loading and resizing, and automatic transition detection. On one NVIDIA L20, SHOT3R with one transition processes 100 frames at 3.106 FPS, compared with 3.224 FPS for HUMAN3R, corresponding to 3.8% additional time. A transition adds one neural evaluation at its first frame. Appendix B reports the controlled breakdown, transition-count study, and separate full-system timing for all seven methods.

5 CONCLUSION

We studied the conflicting roles of recurrent history at shot transitions: pre-transition history provides a reference for relating successive shots, yet propagating the same state can induce time-varying drift within the new shot, while reinitialization alone loses the cross-shot reference. SHOT3R uses history only in a temporary alignment path, while an independently reinitialized state drives subsequent recurrence. Geometry-based association connects identities, and a shared transform maps camera, human, and scene predictions into the common world frame.

Controlled experiments support treating alignment and state propagation as separate decisions. Across all three benchmarks, SHOT3R improves WA-MPJPE, camera trajectory accuracy, and IDF1 over continuous-state recurrence; its gains are most pronounced under large viewpoint changes. These improvements require neither future frames nor per-video optimization and add only 3.8% time over HUMAN3R in the standardized one-transition reconstruction test. More broadly, our results suggest that recurrent history can serve as a temporary coordinate reference for inter-shot alignment without being propagated as the recurrent state of the new shot. Complex edits, repeated entrances and exits, and rapid visual changes remain open challenges.

AI USE STATEMENT

Language-model assistance was used for planning, prose editing, analysis support, code scaffolding, and editable vector layout and annotation. It did not synthesize evaluation inputs, reconstruction panels, ground truth, or numerical measurements. The authors inspected the underlying records and are responsible for every protocol, result, figure, citation, and claim.

ETHICS STATEMENT

The human-capture datasets are public or research-licensed and are used under their providers' consent, access, and research-use conditions. We do not infer participant identity, link participants across datasets, or use the material for surveillance. Restricted imagery, annotations, and checkpoints remain in access-controlled storage and are not redistributed when prohibited by their licences.

REPRODUCIBILITY STATEMENT

We retain fixed evaluation definitions, evaluator code, per-case records, method-to-code mappings, streaming-input checks, a controlled state-isolation experiment with identical weights, an alignment-state-propagation control, viewpoint-clustered statistics, an association-cue ablation, and repeated-transition controls. Tables are generated from machine-readable summaries with source hashes. End-to-end reproduction additionally requires the licensed data and checkpoints; release materials will include code, protocol manifests, aggregate measurements, and non-identifying case identifiers to the extent permitted by those licences.

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Shot3R: Streaming Multi-Person 4D Reconstruction from Multi-Shot Videos by Decoupling Alignment and Recurrent State Propagation

Supplementary Material

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SUPPLEMENTARY CONTENTS

The supplement is organized into three parts. Part I defines the evaluation contract and documents training, inference, detector, runtime, and coordinate conventions. Part II provides the complete comparisons and controlled evidence under that contract. Part III states the supported scope and limitations. Page numbers link to the corresponding sections.

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Highlights and suggested reading. For a compact path through the principal evidence, we recommend:

Method and reproducibility. Section A fixes datasets, metrics, and comparison scope; Sections B–C document trainable parameters, detector, runtime, coordinate transforms, and repeated inference.

State-transition evidence. Section D.4.1 gives the same-checkpoint carry-over control; Sections D.3–D.5 provide the corresponding reconstruction and association analyses.

Viewpoint and repeated-transition evidence. Section D.1 analyzes viewpoint change, and Section D.5.3 verifies that the transition operation can be reapplied on three-shot streams.

Visual and scope evidence. Section D.6 retains the full seven-method qualitative comparison, while Section E states the supported scope and remaining limitations.

Part I: Protocols, Method Details, and Inference

A EVALUATION PROTOCOLS

Datasets. The three main benchmarks are constructed from synchronized multi-camera captures, but are presented to every method as temporally ordered monocular videos: only one view is observed at any time. EgoBody, EgoHumans, and Harmony4D contain 129, 90, and 88 two-shot videos, respectively. Table 3 summarizes the three evaluation sets.

Table 3: Datasets used in the multi-shot multi-person evaluation.

Dataset	Videos	Captures	People	Shots	Aggregation
EgoBody	129	43	Multiple	2	Recording
EgoHumans	90	27	Multiple	2	Video
Harmony4D	88	25	Multiple	2	Video

Table 4: Temporal access, preprocessing, and native output scope of the evaluated public methods.

Access	Method	Preproc.-free	Camera	Scene	Transition
Offline	PromptHMR (SPEC)	×	×	×	×
	JOSH	×	✓	✓	×
	TRAM	×	✓	×	×
Semi-online	OnlineHMR	×	✓	×	×
Online	TRACE	✓	×	×	×
	HUMAN3R	✓	✓	✓	×
	SHOT3R	✓	✓	✓	✓

A check mark reports an available property or native output, not reconstruction quality. Preproc.-free excludes precomputed detection, tracking, camera-frame humans, camera, mask, depth, and contact cues. TRAM is evaluated only in the runtime protocol.

Input and supervision. All methods within a dataset receive the same RGB sequence. Camera calibration, 3D bodies, person identities, and annotated transition locations are used only for evaluation and are not given to the model. The only exceptions are the two controlled intermediate configurations in Table 2, which use annotated transition timing to examine the effects of state reinitialization and alignment under a controlled boundary setting.

Metrics. W-MPJPE (W) estimates a Sim(3) transform from the earliest valid observations and keeps it fixed for subsequent frames; WA-MPJPE (WA) estimates Sim(3) from all valid observations. We additionally report camera absolute trajectory error (ATE), IDF1 for identity consistency, and Coverage for the fraction of ground-truth person instances represented by valid predictions. Because the alignment definitions differ across datasets, absolute errors are compared only within the same dataset.

EgoBody is aggregated within each recording and then averaged over its 43 recordings. EgoHumans and Harmony4D use clip-level means. Geometry errors that require a valid person match are always accompanied by their number of valid clips or by Coverage.

Public methods and offline reference. OnlineHMR, TRACE, and PromptHMR are evaluated with their released implementations on the same RGB sequences as Shot3R. OnlineHMR provides a separate camera trajectory; the released TRACE and PromptHMR outputs are evaluated for human reconstruction under our protocol. Parenthesized counts in Table 1 show the number of clips with valid geometry. OnlineHMR combines semi-online tracking with a global camera backend, whereas PromptHMR processes the complete video; these temporal differences are not treated as equivalent to Shot3R’s streaming inference. JOSH uses its released implementation with Pi3X initialization and one fixed configuration across all three main protocols. It receives each complete RGB sequence offline, but no ground-truth camera, identity, body shape, transition, or shot-wise transform; we do not manually stitch identities. Failures remain in complete-dataset identity and Coverage denominators, and unavailable errors are not imputed. TRAM is run with its released full prediction path for the efficiency study; this profiling run is not included in the three-dataset accuracy comparison.

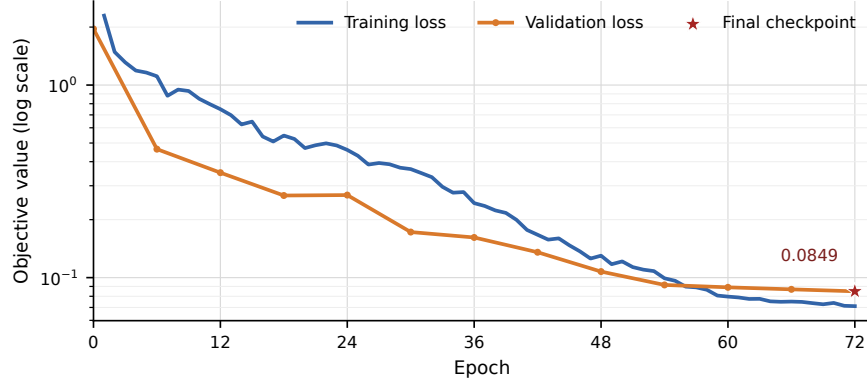


Figure 7: Training and validation objectives of the reported SHOT3R model. The curves use the aggregation rules defined above; the star marks the epoch-72 checkpoint used throughout.

B TRAINING AND IMPLEMENTATION DETAILS

Training configuration. The reported SHOT3R model is initialized directly from pretrained HUMAN3R and trained on the lbn1/zzr subjects of AvatarReX and THuman00/02. Each source contains 8,000 shot-transition clips and 2,000 continuous clips. A shot-transition clip follows $A(t), A(t+1), B(t+2), B(t+3)$, whereas a continuous clip contains four consecutive frames from one camera; both carry camera and human annotations in a shared world frame.

The image encoder, recurrent decoder, point-map head, base camera and human heads, and Multi-HMR backbone (Baradel et al., 2024) are frozen. The trainable shot-alignment module comprises a 14.39M alignment block with three typed tokens, a 1.78M camera latent-residual head, a 2.96M human latent-residual head, and 0.50M rank-8 LoRA parameters in the camera and human heads (Hu et al., 2022): 19.62M parameters in total, or 1.63% of the 1.208B-parameter model. The three alignment tokens occupy the 768-dimensional decoder space, and the alignment gate is shared with the human residual branch.

Each epoch contains 100 updates with ten four-frame sequences per batch; AvatarReX and THuman are sampled with probabilities 0.6 and 0.4. Training runs for 72 epochs and 7,200 optimizer updates. AdamW uses $\beta = (0.9, 0.95)$, weight decay 0.05, four-epoch linear warm-up to 2×10^{-4} , and cosine decay to 10^{-6} , with automatic mixed precision and gradient checkpointing.

Let e and e^{raw} be the corrected and uncorrected camera errors after normalizing translation by 0.5 m and rotation by 45° . We optimize the shot-alignment module with

$$\begin{aligned} \mathcal{L}_{\text{align}} = & \mathcal{L}_t + 5\mathcal{L}_R + 0.05 \text{BCE}(g, \text{clip}(e^{\text{raw}}, 0, 1)) \\ & + 0.05[e - e^{\text{raw}}]_+ + 10^{-5}\mathcal{R}_{\text{cam}} + 10\mathcal{L}_h + 10^{-5}\mathcal{R}_{\text{human}}, \end{aligned} \quad (11)$$

where $\mathcal{L}_t$ and $\mathcal{L}_h$ are Smooth-L1 camera and human translation losses, $\mathcal{L}_R$ is quaternion geodesic distance, g is the learned alignment gate, and the residual terms constrain the camera and human latent updates. Camera and human supervision is evaluated across the four-frame training sequence; no ground truth is used at inference.

Training convergence. Validation is run every six epochs on four fixed splits defined by dataset (AvatarReX or THuman) and clip type (shot transition or continuous). Figure 7 plots the global mean training objective and the mean of the four validation median losses. The epoch-72 checkpoint is used in all reported experiments.

Shot-transition detection. The detector computes 20-dimensional colour, grayscale, hash, edge, optical-flow, histogram, and ORB statistics from consecutive RGB frames and retains the most recent three frame pairs. A single-layer GRU (Cho et al., 2014) with a 32-dimensional hidden state and a linear binary head predicts the transition probability. It is trained for 120 epochs using binary cross-entropy and AdamW with a learning rate of 2×10^{-3} and weight decay of 2×10^{-3} (seed 20260805).

Table 5: Full-system runtime on five fixed 150-frame Harmony4D multi-shot streams (750 frames total).

Category	Method	Camera	Scene	s / 150 frames ↓	FPS ↑
Offline	PromptHMR (SPEC)	×	×	708.7 ± 209.0	0.212
	TRAM	✓	×	455.5 ± 60.6	0.329
	JOSH	✓	✓	949.6 ± 286.0	0.158
Semi-online	OnlineHMR	✓	×	1007.8 ± 124.7	0.149
Online	TRACE	×	×	45.0 ± 21.5	3.334
	HUMAN3R	✓	✓	120.4 ± 3.2	1.246
	SHOT3R	✓	✓	167.8 ± 4.0	0.894

Seconds are mean ± population standard deviation; FPS is 750 divided by total wall time. Camera denotes an independently evaluable physical-camera trajectory, and Scene denotes scene geometry. Timing includes required prediction-process launches and model loading through native prediction save; multi-stage releases report the sum of their required stages. Rendering and evaluation are excluded. Output scope therefore remains relevant when interpreting throughput.

Table 6: Reconstruction time and peak memory for one 100-frame EgoHumans clip on an NVIDIA L20 (resolution 512, batch size 1, FP32).

Method	s / 100 frames ↓	FPS ↑	Peak VRAM (GiB) ↓
Human3R	31.020	3.224	5.292
Shot3R, no transition	31.125	3.213	5.359
Shot3R, one transition	32.191	3.106	5.246

The decision threshold is 0.5. The frozen training ledger contains 408 frame pairs (216 positive) from eight constructed patterns spanning AvatarReX, THuman, and two MVHuman transition settings. In the formal two-shot protocol, the first score above threshold triggers the state change. No future frames, camera calibration, body annotations, or person identities enter the detector.

The first trigger coincides with the annotated boundary in all 129 EgoBody, 90 EgoHumans, and 88 Harmony4D streams. Harmony4D has no off-boundary positive response across 13,200 scored frames.

Full-system runtime. Table 5 reports full-system wall time for seven methods on five fixed 150-frame Harmony4D multi-shot streams. The set was frozen before runtime aggregation and contains hugging, grappling2, sword2, mma4, and mma5; each input comprises two sequential 75-frame monocular shots. Every required prediction stage uses one NVIDIA L20, and timing includes its fresh process launch and model loading through native prediction serialization. For releases with multiple required stages, we sum their stage times.

The scope includes model construction and checkpoint loading, plus any detection, segmentation, tracking, camera recovery, temporal/global optimization, and human recovery required by the released prediction path. The retained PromptHMR and TRACE runs also include lightweight per-case hardlink staging, without image decoding or re-encoding. Dataset download and archive extraction, metric evaluation, paper-format conversion, and rendering are excluded. FPS is the micro-average $750 / \sum_i t_i$; seconds report the mean and population standard deviation across the five clips. All 35 method-clip runs completed successfully; a failed run would not have been silently omitted.

Controlled reconstruction cost. Table 6 separately measures the controlled reconstruction component on one 100-frame EgoHumans clip. Timing includes model inference, output decoding, person association, and world-coordinate alignment, but excludes image loading, resizing, and transition detection. It isolates the cost of the transition operation and is not mixed with the full-system FPS in Table 5. A single transition adds 1.066 s (3.42%) relative to Shot3R without a transition.

Transition-count control. To isolate the number of shot boundaries, we duplicate one fixed 150-frame EgoBody RGB input to form the same 300-output-frame timeline for every configuration and place 0, 1, 3, or 5 annotated transitions. Each boundary requires one additional evaluation of its

Table 7: Controlled SHOT3R runtime and peak allocation versus the number of transitions, with 300 output frames. Peak is the maximum PyTorch allocated memory over the three measured runs.

Transitions	Neural evaluations	Median time (s)	Output FPS	Peak (GiB)
0	300	105.423	2.846	5.82
1	301	107.286	2.796	5.48
3	303	103.466	2.900	5.30
5	305	102.585	2.924	5.25

boundary frame, so C transitions execute exactly $300 + C$ neural frames. Hardware, precision, and timing scope match the controlled reconstruction test above; each configuration has one warm-up followed by three measured repetitions, from which we report the median.

Median time spans 102.6–107.3 s and does not increase monotonically with the number of transitions; peak allocation does not increase either. Thus, within this test range, up to five boundary operations do not produce a throughput change distinguishable from the measured execution variation. Because more transitions also split inference into shorter segments, the lower peaks must not be interpreted as constant full-system memory for arbitrarily long sequences.

C ADDITIONAL METHOD DETAILS

Shot-alignment module. At the first frame of a new shot, the history-conditioned alignment path inserts three tokens after the backbone camera token. A learned query attends to the projected current image tokens, the camera token, and available human tokens to obtain c_b . A second query attention pools the recurrent state tokens, camera memory, and preceding decoder-refined alignment tokens into m_{b-1} . The semantic mixing weight in Eq. 4 is $\alpha_b = \sigma(\text{MLP}_{\text{sem}}[c_b, m_{b-1}])$.

Here $\Delta p_b = p_b - p_{b-1}$; $\bar{z}_{b-1}$, δ_{b-1} , and g_{b-1} in Eq. 4 are the preceding alignment summary, camera correction, and gate.

The camera-alignment token is produced from the current camera token, the preceding one, their latent difference, and m_{b-1} . The temporal-context token pools the preceding decoder-refined alignment tokens, camera residual, and residual gate. A learned type embedding distinguishes the three roles; LayerNorm is applied before decoder attention refines them jointly with the backbone image, camera, and human tokens.

A learned residual updates the temporary camera representation before it is decoded by the frozen camera head. The human residual branch supplies additional supervision during training. Once the inter-shot transform has been estimated, the temporary predictions and state are discarded; only the reinitialized state is propagated through the new shot.

Person association. For a pre-transition person i and post-transition person j , the matching cost combines root position, torso orientation, and root-centered joint configuration:

$$d_{ij} = \hat{d}_{\text{root}}(i, j) + \hat{d}_{\text{torso}}(i, j) + \hat{d}_{\text{joint}}(i, j). \quad (12)$$

Each term is normalized by the median of its valid pairwise values. Hungarian assignment returns $\min(N^-, N^+)$ one-to-one pairs, and no confidence threshold is applied. Later frames retain the identities initialized at the first post-transition frame. The matching uses predictions only, without ground-truth identity, calibration, or future frames.

Shared camera–human transform. The inter-shot camera transform first maps the new shot into the existing world frame. The median displacement of matched people then provides the shared translation in Eq. 9. The resulting transform is applied equally to camera poses, joints, and mesh vertices. Unmatched new-shot detections receive new anonymous identities. If no pair is available, the shared translation is omitted and only camera alignment is retained.

Coordinate convention. All camera poses are homogeneous camera-to-world transforms $C = \begin{bmatrix} R & t \\ \mathbf{0}^\top & 1 \end{bmatrix}$. A world transform T is left-composed with the camera, world-space scene points, and human geometry: $C \leftarrow TC$, $P^w \leftarrow TP^w$, and $X \leftarrow TX$, with all points expressed in homogeneous coordinates. The alignment module does not deform the camera-local scene geometry P^c ; its world-space realization is obtained from the corrected camera pose, $P^w = CP^c$, and is therefore transformed consistently with the camera and human predictions.

Multiple transitions. At every later transition, the current recurrent state is used only to estimate the new spatial relation, while a newly initialized state supplies subsequent recurrence. Previously emitted frames remain unchanged. For $b_1 < b_2$, the first transform affects frames from b_1 onward, and the second affects only frames from b_2 onward; later predictions use the composed transform T_2T_1 . Section D.5.3 evaluates this composition on three-shot sequences.

Part II: Additional Experimental Evidence

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D ADDITIONAL EXPERIMENTS

This section follows the main questions of the paper: performance under large viewpoint changes, comparison with public methods, repeated shot transitions, and detailed results on the three multi-person benchmarks. Tables already reported in the main paper are not repeated.

D.1 LARGE VIEWPOINT CHANGES

The main paper reports the dataset-balanced summary. Table 8 further gives the results on the predefined farthest-view subset of each dataset. The subsets contain 43 EgoBody recordings, 22 EgoHumans captures, and 25 Harmony4D captures and are defined without reference to method performance.

To test whether the advantage is more pronounced at large viewpoint changes, we compare the paired gain in the large-viewpoint subset with the gain on the remaining views. Positive values in Table 9 indicate a larger Shot3R gain under larger viewpoint changes. The dataset-balanced intervals are positive for W, WA, and IDF1, while individual datasets need not improve on every metric.

To examine the camera-level relation directly, we use the exact camera angle for all 90 EgoHumans cases and correlate viewpoint change with the Shot3R gain over HUMAN3R. Intervals use a two-stage bootstrap that first resamples the 27 capture clusters and then cases within each selected capture (Davison & Hinkley, 1997).

Continuous-angle analysis supports a stronger relationship for camera and WA errors, consistent with SHOT3R acting on the spatial relation at a shot transition, but does not establish monotonic improvement for every human or identity metric. The stratified results retain the degradations observed in the small and medium ranges.

Table 8: Results on the predefined large-viewpoint subsets. W/WA are in mm and camera ATE is in m (Sim(3) for EgoBody/Harmony4D and SE(3) for EgoHumans).

Dataset	Method	W↓	WA↓	Camera↓	IDF1↑	Cov.↑
EgoBody	HUMAN3R	505.1	321.8	0.253	0.758	0.974
	SHOT3R	354.3 (29.9%↓)	273.3 (15.1%↓)	0.016	0.983	0.974
EgoHumans	HUMAN3R	1228.4	400.0	3.390	0.404	0.611
	SHOT3R	886.1 (27.9%↓)	315.9 (21.0%↓)	2.326	0.559	0.576
Harmony4D	HUMAN3R	550.2	325.7	0.250	0.418	0.866
	SHOT3R	552.6 (0.4%↑)	260.7 (20.0%↓)	0.017	0.622	0.859

Table 9: Additional gain at large viewpoints relative to the remaining views, with 95% recording/capture-cluster bootstrap intervals.

Dataset	W (mm)	WA (mm)	IDF1
EgoBody	+132.8 [+10.3, +286.1]	+25.2 [−12.9, +68.6]	+0.166 [+0.109, +0.224]
EgoHumans	+270.1 [−13.7, +611.8]	+73.2 [+7.0, +144.9]	+0.072 [−0.010, +0.160]
Harmony4D	+28.7 [−39.3, +102.4]	+73.2 [+35.1, +114.5]	+0.169 [+0.083, +0.253]
Dataset-balanced	+143.9 [+36.1, +269.5]	+57.2 [+28.6, +87.5]	+0.136 [+0.092, +0.180]

Table 10: Relation between viewpoint change and reconstruction gain on EgoHumans. Left: case-level Spearman correlation with two-stage capture-cluster bootstrap intervals. Right: mean gains in the predefined angle strata. W/WA are in mm, camera rotation in degrees, and camera translation in m; positive values favour SHOT3R.

(a) Continuous-angle correlation			(b) Mean gain by predefined angle stratum					
Gain quantity	ρ	95% CI	Stratum	W	WA	Cam. R	Cam. T	IDF1
W-MPJPE	0.266	[−0.011, 0.510]	Small	−83.1	−25.7	−0.11	−0.306	0.034
WA-MPJPE	0.287	[0.021, 0.519]	Medium	23.5	4.4	−1.02	−0.984	0.159
Boundary camera rotation	0.413	[0.126, 0.668]	Large	290.5	57.5	10.93	0.832	0.057
Boundary camera translation	0.519	[0.300, 0.684]	Extreme	342.4	84.1	37.26	2.542	0.154
IDF1	0.104	[−0.174, 0.368]						

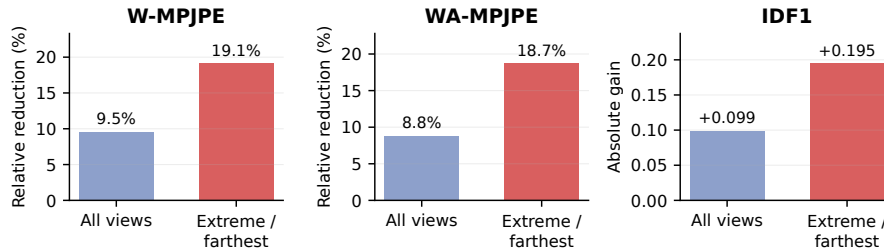


Figure 8: Dataset-balanced improvements over HUMAN3R on all views and on the predefined large-viewpoint subsets. W/WA show relative reductions; IDF1 shows absolute gain.

D.2 COMPARISON WITH PUBLIC METHODS

Table 1 reports every executed method on the same EgoBody and EgoHumans RGB sequences, while Table 11 gives the complete Harmony4D comparison. We additionally compare Shot3R and OnlineHMR on the clips for which OnlineHMR provides valid geometry; IDF1 and Coverage are computed over all sequences. Table 14 shows lower human and camera errors for Shot3R on all three datasets. All intervals exclude zero except WA on EgoHumans. OnlineHMR uses a semi-online tracker and a separate global-camera backend, so this comparison does not imply identical temporal access.

Beyond the three primary protocols, four Harmony4D three-shot streams compare repeated transition handling in Shot3R, Human3R, and OnlineHMR; Section D.5.3 reports their complete results.

The official JOSH release is run on all 307 streams from the three primary protocols. Inference and coordinate checks complete for 302 streams; W/WA are evaluable on 292 and camera ATE on 302. Two EgoBody streams fail because contact-point initialization returns an empty set, and three EgoHumans streams fail because a shot-boundary resolution change is incompatible with the released SAM3 mask pipeline; all 88 Harmony4D streams succeed. Ten otherwise completed EgoHumans streams lack the initial human match required by the fixed W/WA evaluation protocol. Table 12 distinguishes failed runs from completed runs that cannot be evaluated by a particular metric.

To remove ambiguity from conditional means computed on different supports, Table 13 restricts each metric to streams evaluable by both Shot3R and JOSH. Shot3R obtains lower W, WA, and camera ATE on all three common-support sets while processing frames without future access. IDF1 and Coverage retain the complete protocol denominator: JOSH is higher on both in EgoHumans, whereas Harmony4D combines higher Shot3R IDF1 with higher JOSH Coverage.

For some of the closest published methods, the released resources are insufficient to evaluate our fixed two-shot sequences under the same protocol. Table 15 therefore distinguishes task scope, temporal access, and whether a release provides the outputs required by our evaluation.

Table 11: Complete comparison on the Harmony4D multi-shot multi-person protocol ($N = 88$).

Category	Method	W ↓	WA ↓	ATE-Sim3 ↓	IDF1 ↑	Coverage ↑
Offline	PromptHMR (SPEC)	963.2 (5)	306.9 (26)	–	0.030	0.034
	JOSH	770.2 (88)	369.7 (88)	0.153 (88)	0.551	0.959
Semi-online	OnlineHMR	750.8 (30)	335.9 (30)	1.364 (88)	0.068	0.159
Online	TRACE	677.1 (4)	270.2 (26)	–	0.016	0.016
	HUMAN3R	496.6	259.6	0.1420	0.553	0.915
	SHOT3R	519.6	247.0	0.0168	0.636	0.913

W/WA are in mm and ATE is in m. Parentheses give the number of streams evaluable for each conditional geometry metric; IDF1 and Coverage use all 88 fixed streams. Geometry means with different support do not form a common ranking.

Table 12: JOSH execution and evaluable cases on the three primary protocols. Failed runs remain in the IDF1/Coverage calculations over all sequences; unavailable human or camera errors are omitted rather than set to zero.

Dataset	Total	Completed	W/WA evaluable	ATE evaluable	Failed
EgoBody	129	127	127	127	2 contact initialization
EgoHumans	90	87	77	87	3 mask-size mismatch [†]
Harmony4D	88	88	88	88	None
Total	307	302	292	302	5

[†]Ten additional completed streams lack the initial match required by W/WA.

Table 13: Case-macro Shot3R–JOSH comparison restricted to streams evaluable by both methods. W/WA are in mm; ATE is Sim(3)-aligned for EgoBody and Harmony4D and SE(3)-aligned for EgoHumans, in m.

Dataset	N	W-MPJPE		WA-MPJPE		ATE		
		Shot3R	JOSH	Shot3R	JOSH	N	Shot3R	JOSH
EgoBody	127	260.2	402.0	202.4	324.9	127	0.017	0.088
EgoHumans	77	887.2	1410.7	337.7	481.3	87	1.289	1.755
Harmony4D	88	519.6	770.2	247.0	369.7	88	0.017	0.154

Table 14: Paired Shot3R gain over OnlineHMR with 95% clustered bootstrap intervals. Positive values favour Shot3R.

Dataset	W gain [95% CI]	WA gain [95% CI]	ATE* gain [95% CI]	IDF1 gain [95% CI]	Coverage gain [95% CI]
EgoBody	730.1 [522.9, 941.6]	233.9 [158.8, 307.9]	0.243 [0.153, 0.348]	0.610 [0.527, 0.691]	0.557 [0.465, 0.647]
EgoHumans	430.3 [128.9, 714.2]	71.5 [-1.3, 139.5]	2.447 [1.906, 3.080]	0.429 [0.369, 0.486]	0.405 [0.332, 0.474]
Harmony4D	340.8 [165.9, 509.2]	134.0 [78.5, 190.3]	1.348 [1.227, 1.463]	0.568 [0.491, 0.640]	0.754 [0.654, 0.835]

Table 15: Task scope and released-resource compatibility for related methods not included in our numerical comparison on identical sequences. ✓ denotes an available capability, ✗ denotes an unavailable capability, and △ denotes a released resource not yet validated under our protocol; † denotes a capability reported in the original paper but not evaluable under our protocol.

Work	Shot transitions	Multi-person	Physical cam.	Shared world frame	Evaluable here	Input / temporal access	Role here
HumanMM (Zhang et al., 2025b)	✓	✗	✓†	✓†	✗	Full-sequence processing	Related work
Multi-THuMBS (On et al., 2026)	✓	✗	✓†	✓†	✗	Offline refinement	Related work
Multishot (Pavlakos et al., 2022b)	✓	✗	✗	✗	✗	Offline optimization	Related work
SLAHMR (Ye et al., 2023)	✗	✓	✓	✗	✗	Offline optimization	Different setting
WHAC / GVHMR (Yin et al., 2024; Shen et al., 2024)	✗	✗	✗	✗	✗	Full-video inference	Single-person setting
Yang et al. (Yang et al., 2026)	✗	✓	✓	✓	✗	Joint multi-view set	Complementary multi-view

Evaluation under our protocol requires human tracks, a camera trajectory, identity assignments, and a common timeline spanning both shots, without ground-truth track selection, manual identity stitching, or a separately fitted post-transition transform. SLAHMR’s released configuration processes one selected shot; the released Multishot demo requires a specified person track. Physical-camera support for SLAHMR is confined to its selected shot. JOSH is no longer listed here because its released full-video implementation is evaluated in Table 1.

D.3 EGOBODY

The main paper reports the overall EgoBody results and principal mechanism comparison. Here we provide local and boundary measures for the controlled alignment–state-propagation comparison, paired confidence intervals, cross-shot continuity, and viewpoint-stratified results.

The two aligned controls use the same checkpoint, annotated boundary, and history-conditioned alignment, disable explicit association and shared translation, and differ only in the state propagated afterward. Their equal first-post camera errors verify that the boundary alignment is held fixed. Local human accuracy is nearly unchanged; reinitialization lowers the first-post root error, whereas continuation lowers the subsequent root error.

The first detector response occurs at the true shot transition in all 129 EgoBody clips, with no early, late, or missed transitions. We report this result in text rather than repeating it in a one-row table.

Table 16: Local and boundary measures when varying the state propagated after alignment on EgoBody. Human errors are in mm, camera rotation in degrees, and the remaining camera/root errors in m.

State propagated	Local human		First post-transition			Post root
	MPJPE	PA-MPJPE	Cam. t	Cam. R	Root	
Continued	71.18	43.80	0.807	8.02	0.798	0.430
Reinitialized	70.94	43.87	0.807	8.02	0.538	0.554

Table 17: Paired Human3R–Shot3R analysis over 43 EgoBody recordings. The interval is a paired bootstrap and positive gain favours Shot3R.

Metric	Mean gain $\uparrow$	95% CI	p -value
W-MPJPE (mm)	62.3	[21.2, 111.3]	0.003708
WA-MPJPE (mm)	31.7	[15.1, 49.2]	0.000639
ATE-Sim3 (m)	0.108	[0.056, 0.173]	$< 10^{-6}$
IDF1	0.114	[0.076, 0.155]	$< 10^{-6}$

Table 18: Local reconstruction and temporal stability on EgoBody. Human errors and acceleration are in mm, ATE-Sim3 in m, and foot sliding in cm.

Method	MPJPE	PA-MPJPE	MPVPE	ATE-Sim3	Accel.	Foot sliding
HUMAN3R	71.2	43.8	79.3	0.215	34.3	2.6
SHOT3R	70.9	43.9	79.0	0.207	28.0	2.2

Table 19: Camera and human continuity at EgoBody shot transitions. Translation and root errors are in m; rotation errors are in degrees.

Method	Boundary cam. t	Boundary cam. R	Inter-shot cam. t	Inter-shot cam. R	Cross-shot root
HUMAN3R	0.840	12.3	0.703	6.3	0.524
SHOT3R	0.679	8.0	0.501	2.4	0.309

Table 20: EgoBody results by viewpoint change. W/WA are in mm and camera error is in m.

Viewpoint change	Method	W	WA	Camera	IDF1
Small	HUMAN3R	223.4	182.1	0.048	0.965
	SHOT3R	201.4	160.6	0.017	0.990
Medium	HUMAN3R	234.9	196.2	0.072	0.888
	SHOT3R	220.8	171.2	0.017	0.981
Extreme	HUMAN3R	505.1	321.8	0.253	0.758
	SHOT3R	354.3	273.3	0.016	0.983

D.4 EGOHUMANS

The viewpoint breakdown in Table 21 shows that the main gains arise at large viewpoint changes; not every metric improves in the small-view subset.

D.4.1 RECURRENT-STATE CARRY-OVER

World-frame reconstruction error alone cannot distinguish losing a fixed inter-shot coordinate relation from changing the trajectory within the new shot. We isolate the latter on a fixed 90-case EgoHumans protocol using the same original HUMAN3R checkpoint. The continuous setting processes both shots in order and carries state across the transition; the reinitialized setting processes the identical 50 frames after the transition from the initial state. Both use the same preprocessing, and no SHOT3R module is enabled.

Each EgoHumans shot is recorded by a static camera, so the ground-truth relative camera motion within the second shot is the identity. We anchor the first prediction after the transition and measure

$$\begin{aligned}\Delta\hat{C}_k &= \hat{C}_{B0}^{-1}\hat{C}_{Bk}, \\ d_T(k) &= \left\| \text{trans}(\Delta\hat{C}_k) \right\|_2, \\ d_R(k) &= \cos^{-1} \left(\frac{\text{tr}(\text{Rot}(\Delta\hat{C}_k)) - 1}{2} \right).\end{aligned}\tag{13}$$

Any fixed world-frame transform between shots cancels in this relative pose, so the measure does not penalize reinitialization for lacking the inter-shot coordinate relation. We average offsets $k = 1, \dots, 16$ within each case, then average cases within a capture and give all 27 captures equal weight.

As Table 23 shows, continued state produces 1.124 m/14.58° mean translation/rotation drift, compared with 0.026 m/0.20° after reinitialization. Reinitialization is lower on all 27 captures for both measures. The capture-paired mean reductions and 95% bootstrap intervals are 1.098 [0.832, 1.372] m and 14.38° [8.92°, 20.49°]; both one-sided paired Wilcoxon tests give $p = 7.45 \times 10^{-9}$.

The conclusion is stable to the measurement window. For windows of 10/32/49 frames, continued versus reinitialized translation drift is 1.045/0.019, 1.256/0.039, and 1.387/0.047 m; rotation drift is 14.34°/0.16°, 15.12°/0.27°, and 15.65°/0.34°. The ordering is also unchanged in all four fixed viewpoint strata. This control does not claim that reinitialization alone solves multi-shot reconstruction—it loses the inter-shot coordinate relation. Instead, it shows that the preceding-shot state causes time-varying error within the subsequent shot, supporting separate treatment of alignment and recurrent-state propagation.

The ablation in Table 24 confirms the same pattern as EgoBody. Reinitialization alone breaks world-frame continuity; camera alignment restores the inter-shot camera relation, while the full method also uses person association and the shared transform to improve human trajectories and identity continuity.

Table 21: EgoHumans results by viewpoint change. W/WA are in mm.

Method	Viewpoint stratum	N	W-MPJPE $\downarrow$	WA-MPJPE $\downarrow$	IDF1 $\uparrow$	Coverage $\uparrow$
Human3R	small	24	767.7	331.5	0.559	0.658
Shot3R	small	24	850.9	357.1	0.593	0.657
Human3R	medium	22	824.7	345.9	0.414	0.637
Shot3R	medium	22	801.2	341.5	0.572	0.605
Human3R	large	22	1221.1	388.6	0.501	0.647
Shot3R	large	22	930.7	331.1	0.558	0.622
Human3R	$\geq 150^\circ$	22	1228.4	400.0	0.404	0.611
Shot3R	$\geq 150^\circ$	22	886.1	315.9	0.559	0.576

The four fixed strata contain 24, 22, 22, and 22 clips. W/WA are available for all clips in this internal comparison; IDF1 and coverage retain the stratum denominator.

Table 22: Paired Human3R–Shot3R analysis on 90 EgoHumans clips from 27 captures. Intervals resample captures and then clips within each capture.

Metric	Mean gain $\uparrow$	95% CI	p -value
W-MPJPE (mm)	138.3	[0.5, 322.9]	0.035713
WA-MPJPE (mm)	28.9	[-6.8, 68.7]	0.057267
ATE-SE3 (m)	0.515	[0.258, 0.925]	$< 10^{-6}$
IDF1	0.100	[0.046, 0.152]	0.000799
Coverage	-0.023	[-0.058, 0.008]	0.122279

Table 23: Isolating recurrent-state carry-over on the fixed 90-case EgoHumans protocol. The same original Human3R checkpoint processes identical frames after the transition with the preceding-shot state or a reinitialized state. We report mean within-shot camera drift over the next 16 frames after anchoring the first pose, with equal weight for each of 27 captures.

State at the shot transition	Mean trans. drift (m) $\downarrow$	Mean rot. drift ($^\circ$) $\downarrow$
Continued	1.1237	14.578
Reinitialized	0.0258	0.200

Table 24: Effects of alignment, state reinitialization, and association on EgoHumans. W/WA/MPJPE are in mm and cross-shot root error is in m.

Method	W	WA	MPJPE	Cross-shot root	IDF1	Coverage
HUMAN3R	1005.1	365.7	107.6	1.723	0.471	0.639
State reinitialization	1248.1	491.3	108.5	2.653	0.455	0.616
Camera alignment	1072.4	370.1	108.5	2.113	0.455	0.616
Camera alignment + association	1058.5	372.6	109.9	2.113	0.556	0.600
SHOT3R	866.8	336.9	108.5	1.227	0.571	0.616

D.5 HARMONY4D

Harmony4D uses the same model and inference settings as the other datasets. The complete method comparison appears in Table 11; Table 25 adds only local body errors and translation error at the shot transition.

D.5.1 SHARED-TRANSLATION WEIGHT

We study the shared-translation weight on 12 Harmony4D training sequences reserved for development, nine of which provide all required metrics. Local MPJPE, MPVPE, IDF1, and Coverage do not change with this output-space weight and are omitted rather than repeated. We use $\lambda = 0.5$ across all datasets. On this development split, it achieves the lowest mean W and WA while retaining competitive camera ATE and cross-shot root continuity, providing a balanced combination of camera- and person-derived alignment.

D.5.2 CROSS-SHOT PERSON ASSOCIATION

Among 88 clips, 84 contain at least one valid person match. Of 147 valid matches, 142 are correct, giving 96.60% pair accuracy and 80.68% identity continuation. The ground-truth-identity row in Table 27 changes only the identity labels and is included as an upper bound.

We further vary only the geometric cues in the matching cost, while retaining the same frozen predictions, per-case median normalization, and Hungarian assignment. Each correspondence is fixed before ground-truth identities are opened and used only for scoring.

Pelvis position already gives high pair accuracy. Adding torso orientation and root-centered joint structure further increases the number of correct pairs, identity continuation, and IDF1. This supports the complementarity of the three predicted geometric cues, while association remains a lightweight supporting mechanism after inter-shot alignment rather than the central representation-learning contribution.

D.5.3 THREE-SHOT CONTROL

Four additional Harmony4D sequences contain three shots and two transitions. Shot3R improves W, WA, IDF1, and cross-shot human continuity, while ATE-Sim3 is slightly higher than Human3R. OnlineHMR completes inference on all four streams, but its 0.121 Coverage leaves only three with evaluable W/WA. Its lower W/WA must therefore be read together with this count; across all four streams, SHOT3R has lower camera error and higher IDF1 and Coverage. These four sequences provide a limited repeated-application check rather than a benchmark-scale evaluation.

Table 25: Additional diagnostics on 88 Harmony4D clips. Local body errors are in mm and transition-translation error is in m.

Method	MPJPE ↓	MPVPE ↓	Boundary trans. ↓
HUMAN3R	98.3	116.7	0.864
SHOT3R (ours)	97.9	116.1	1.042

Table 26: Sensitivity to the shared-translation weight on the Harmony4D development split. W/WA are in mm; ATE and cross-shot root error are in m.

λ	W mean ↓	WA mean ↓	ATE ↓	Cross-shot root ↓
0.00	630.4	322.7	0.020	1.175
0.25	536.2	299.1	0.019	0.857
0.50 (used)	499.4	281.5	0.017	0.585
0.75	517.9	288.2	0.016	0.507
1.00	606.5	321.5	0.015	0.560

Table 27: Cross-shot person association on Harmony4D. Pair accuracy uses 147 evaluator-valid endpoint pairs, whereas identity continuation uses all 176 ground-truth continuations. The ground-truth-identity row changes only identity labels and serves as an evaluation upper bound.

Measure	Result
Clips with at least one valid match	84 / 88
Pair accuracy	142 / 147 (96.60%)
Clip-level accuracy (95% CI)	94.05% [88.10, 98.81]
Identity continuation	142 / 176 (80.68%)
Framewise IDF1 (95% CI)	63.61% [59.33, 67.99]
Ground-truth identity upper bound (95% CI)	69.41% [65.76, 73.01]

Table 28: Association-cue ablation on 88 Harmony4D cases. Pair accuracy uses evaluator-valid endpoint matches for each configuration, identity continuation uses all 176 cross-shot ground-truth continuations, and IDF1 is averaged over the 88 cases.

Association cues	Correct / evaluable	Pair accuracy	Identity continuation	IDF1
Pelvis	138 / 144	95.83%	78.41%	0.620
Pelvis + torso	139 / 144	96.53%	78.98%	0.630
Pelvis + torso + root-centered joints	142 / 147	96.60%	80.68%	0.635

Table 29: Harmony4D three-shot results. W/WA are in mm; camera and root errors are in m. Parentheses give the number of evaluable sequences; Coverage is computed over all four streams.

Method	W	WA	ATE-Sim3	IDF1	Coverage	Cross-shot root
HUMAN3R	2091.9	719.4	1.618	0.377	0.936	4.539
SHOT3R	1304.8	651.2	1.700	0.434	0.936	1.725
OnlineHMR	1065.7 (3)	304.3 (3)	3.393	0.055	0.121	—

D.6 FULL QUALITATIVE COMPARISON

The main paper enlarges the four comparisons most directly related to our claim. Figure 9 retains all seven evaluated methods on the same three transitions. Each column shows the method’s available native output; unavailable camera or scene geometry is not imputed.

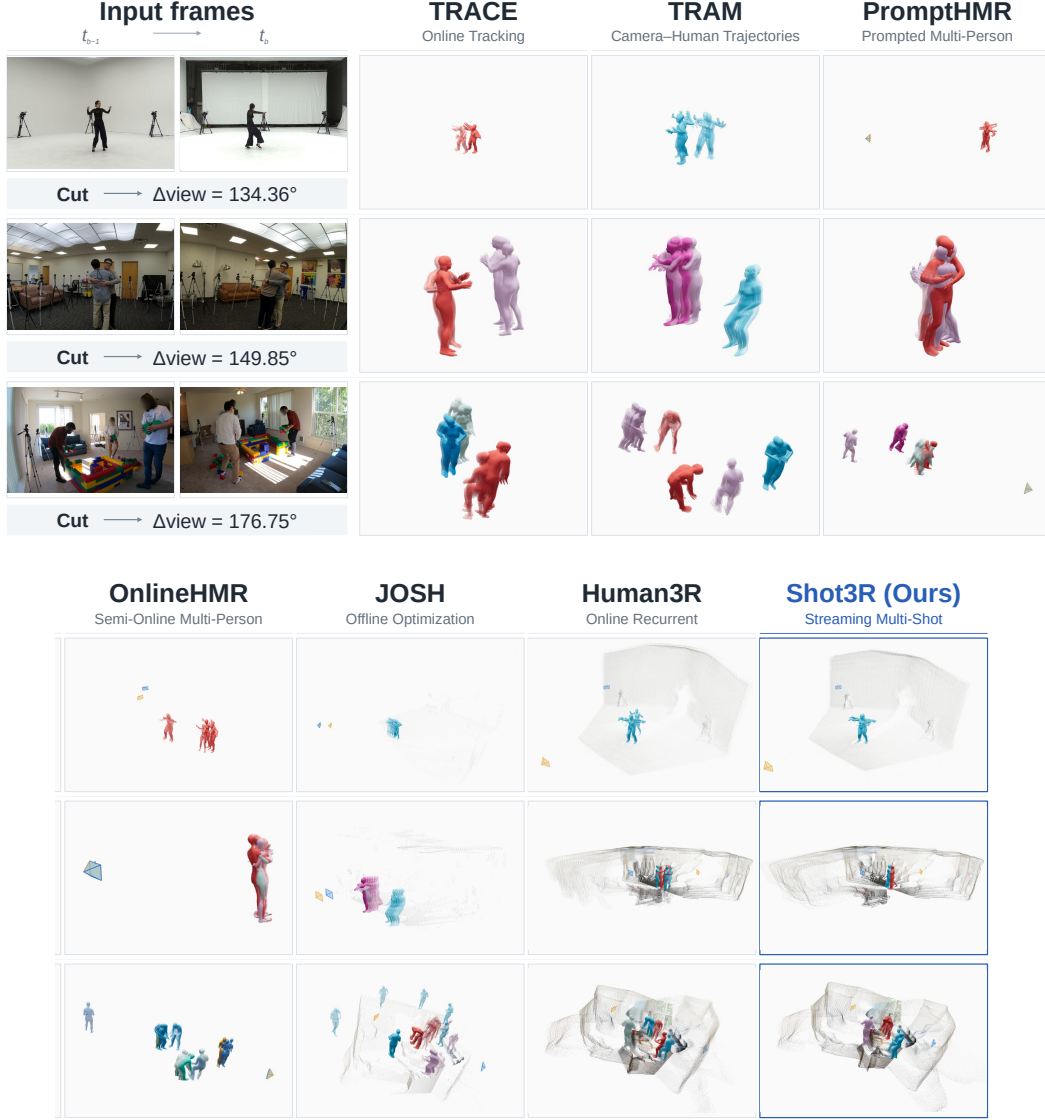


Figure 9: **Complete qualitative comparison.** TRACE, TRAM, PromptHMR, OnlineHMR, JOSH, HUMAN3R, and SHOT3R are shown on the same inputs. Viewpoint changes are 134.36° , 149.85° , and 176.75° from top to bottom; blue borders mark SHOT3R.

Part III: Scope and Limitations

E LIMITATIONS

Our evaluation focuses on shot transitions caused by viewpoint changes within the same physical scene. Maintaining spatial and identity consistency under complex editing, repeated entrances and exits, rapid visual changes, and very long videos remains challenging.

Streaming and offline methods have different temporal access, which is stated throughout the comparisons. Human errors computed only on valid matches should also be read together with Coverage. Improved camera or identity consistency does not imply improvement in every body metric; the complete Harmony4D results retain these metric-level differences.